BGIseq Whole Genome Sequencing Report

2018/4/10



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Results

1 Data Production

To discover genetic variations in this project, we performed whole genome sequencing of 7 DNA sample(s) with averagely 141,314.48 Mb raw bases. After removing low-quality reads we obtained averagely 1,405,040,646 clean reads (140,504.06 Mb). The clean reads of each sample had high Q20 and Q30 , which showed high sequencing quality. The average GC content was 41.50%. All whole genome sequencing data production was summarized in Table1. The base quality scores on clean reads per sample were plotted(Figure1).

Table 1 Summary of whole genome sequencing data (Download)

Samples	Raw reads	Raw bases (Mb)	Clean reads	Clean bases (Mb)	Clean data rate (%)	Clean read Q20 (%)	Clean read Q30 (%)	GC content (%)
622.5.041	1,440,666,050	144,066.61	1,435,387,646	143,538.76	99.63	96.63	88.31	41.37
752.33	1,449,096,150	144,909.61	1,442,985,680	144,298.57	99.58	96.97	88.90	41.65
799.36	1,426,535,604	142,653.56	1,414,254,240	141,425.42	99.14	96.94	88.80	41.61
805.044	1,395,573,416	139,557.34	1,391,252,932	139,125.29	99.69	96.42	87.75	41.52
814.292	1,402,974,662	140,297.47	1,396,865,294	139,686.53	99.56	96.55	87.99	41.31
E1	1,405,062,414	140,506.24	1,398,898,462	139,889.85	99.56	96.19	87.16	41.37
E3	1,372,105,084	137,210.51	1,355,640,274	135,564.03	98.80	96.16	86.83	41.66
Average	1,413,144,768	141,314.48	1,405,040,646	140,504.06	99.42	96.55	87.96	41.50

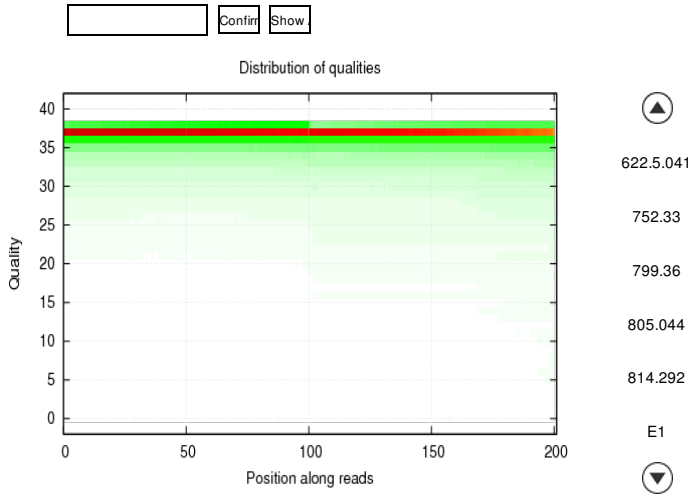


Figure 1 Distribution of base quality scores on clean reads.

X-axis is positions along reads. Y-axis is quality value. Each dot in the image represents the quality score of the corresponding position along reads.

2 Summary Statistics of Alignment

Total clean reads per sample were aligned to the mouse reference genome (mm10) using Burrows-Wheeler Aligner (BWA). On average, 99.69% mapped successfully and 87.75% mapped uniquely. The duplicate reads were removed from total mapped reads, resulting in about 6.68%

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duplicate rate and 46.89-fold mean sequencing depth on the whole genome excluding gap regions. On average per sequencing individual, 94.61% of the whole genome excluding gap regions were covered by at least 1X coverage, 94.13% had at least 4X coverage and 93.26% had at least 10X coverage(Table2). In addition, the distributions of per-base sequencing depth and cumulative sequencing depth were shown as Figure2 and Figure3, respectively. The insert size distribution of paired sequencing reads was plotted in Figure4.

Table 2 Summary statistics of alignment （Download)

Samples	Clean reads	Clean bases (Mb)	Mapping rate (%)	Unique rate (%)	Duplicate rate (%)	Mismatch rate (%)	Average sequencing depth (X)	Coverage (%)	Coverage at least 4X (%)
622.5.041	1,435,387,646	143,538.76	99.68	87.75	6.80	0.41	47.86	93.80	93.59
752.33	1,442,985,680	144,298.57	99.70	88.89	5.50	0.36	48.84	96.54	95.49
799.36	1,414,254,240	141,425.42	99.70	87.75	6.74	0.41	47.17	93.79	93.55
805.044	1,391,252,932	139,125.29	99.70	87.98	6.61	0.44	46.47	93.80	93.59
814.292	1,396,865,294	139,686.53	99.71	87.38	7.18	0.43	46.33	93.79	93.56
E1	1,398,016,326	139,801.63	99.61	87.45	6.81	0.45	46.56	96.72	95.54
E3	1,355,640,274	135,564.03	99.71	87.08	7.11	0.42	44.98	93.82	93.60
Average	1,404,914,627	140,491.46	99.69	87.75	6.68	0.42	46.89	94.61	94.13

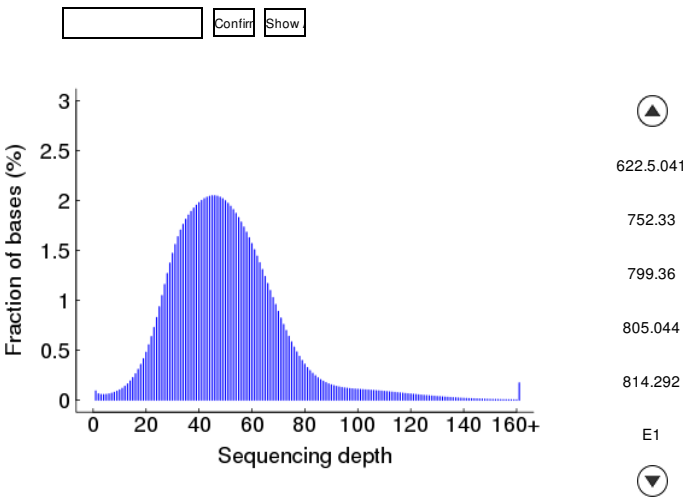


Figure 2 The distribution of per-base sequencing depth on the whole genome.

X-axis denotes sequencing depth, while Y-axis indicates the percentage of the whole genome excluding gap regions under a given sequencing depth.

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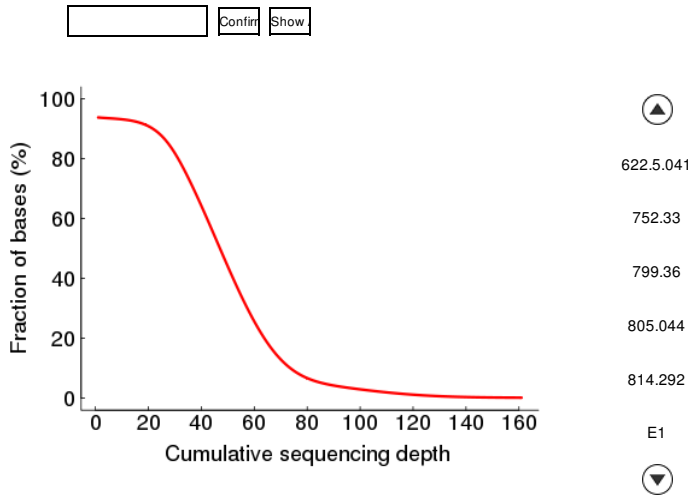


Figure 3 Cumulative depth distribution on the whole genome.

X-axis denotes sequencing depth, and Y-axis indicates the fraction of the whole genome excluding gap regions that achieves at or above a given sequencing depth.

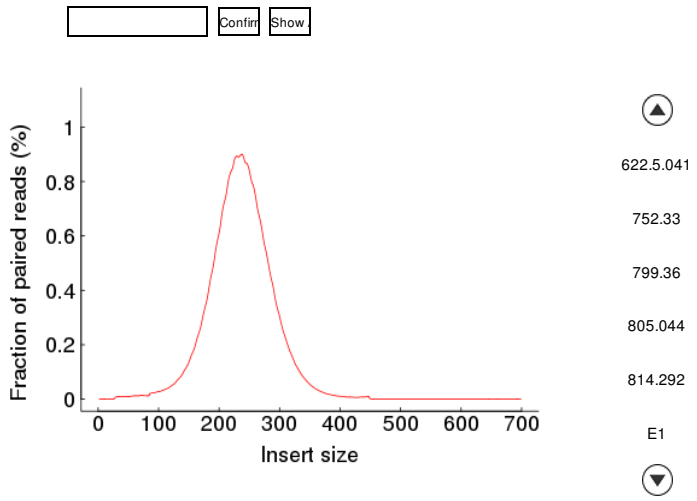


Figure 4 Insert size distribution of paired reads.

X-axis denotes insert size of paired reads, and Y-axis shows the fraction of paired reads with a given insert size.

3 Data Quality Control

The strict data quality control (QC) was performed in the whole analysis pipeline for the clean data , the mapping data, the variant calling, etc. Several quality control items for each sample were checked in Table3, where 'Y' showed PASS and 'N' showed FAIL. If some criteria were not met, measures such as re-sequencing or other effective methods would be carried out to improve the data quality and ensure qualified sequencing data.

Table 3 Data quality control for samples （Download)

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Samples	Clean read Q20 (%)	Clean read Q30 (%)	GC content (%)	Mapping rate (%)	Duplicate rate (%)	Mismatch rate (%)	Average sequencing depth (X)	Coverage (%)	Coverage at least 4X (%)
622.5.041	Y(96.63)	Y(88.31)	Y(41.37)	Y(99.68)	Y(6.80)	Y(0.41)	Y(47.86)	N(93.80)	N(93.59)
752.33	Y(96.97)	Y(88.90)	Y(41.65)	Y(99.70)	Y(5.50)	Y(0.36)	Y(48.84)	N(96.54)	Y(95.49)
799.36	Y(96.94)	Y(88.80)	Y(41.61)	Y(99.70)	Y(6.74)	Y(0.41)	Y(47.17)	N(93.79)	N(93.55)
805.044	Y(96.42)	Y(87.75)	Y(41.52)	Y(99.70)	Y(6.61)	Y(0.44)	Y(46.47)	N(93.80)	N(93.59)
814.292	Y(96.55)	Y(87.99)	Y(41.31)	Y(99.71)	Y(7.18)	Y(0.43)	Y(46.33)	N(93.79)	N(93.56)
E1	Y(96.19)	Y(87.16)	Y(41.37)	Y(99.61)	Y(6.81)	Y(0.45)	Y(46.56)	N(96.72)	Y(95.54)
E3	Y(96.16)	Y(86.83)	Y(41.66)	Y(99.71)	Y(7.11)	Y(0.42)	Y(44.98)	N(93.82)	N(93.60)

4 SNP Results

Overall, we identified 3,790,496 SNPs in all individuals. Of these variants, 80.81% were represented in dbSNP and 727412.00% were annotated in the 1000 Genomes Project database. The number of novel SNPs was NA. The ratio of transition to transversion (Ti/Tv) was 1.87. Of overall SNPs , 17,928 were synonymous, 11,657 were missense, 35 were stoploss, 184 were stopgain, 11 were startloss and 158 were splice site. The Ti/Tv of coding SNPs was 2.82 (Table5). The summary statistics of SNPs was shown in Table4.

Table 4 Summary statistics for identified SNPs （Download)

Samples	Total SNPs	Fraction of SNPs in dbSNP (%)	Novel	Homozygous	Heterozygous	Intronic	5' UTRs	3' UTRs	Upstream	Do
622.5.041	3,602,010	83.88	580,815	2,024,535	1,577,475	1,220,765	3,884	23,315	56,533	57,3
752.33	3,547,667	82.58	618,164	2,388,525	1,159,142	1,199,051	3,922	23,140	55,983	56,4
799.36	3,447,474	84.25	543,082	2,088,455	1,359,019	1,170,843	3,739	22,459	54,803	55,3
805.044	3,602,485	83.78	584,250	2,025,461	1,577,024	1,221,070	3,868	23,257	56,573	57,4
814.292	3,575,271	84.69	547,457	2,030,492	1,544,779	1,210,267	3,822	23,073	56,124	56,7
E1	3,422,870	87.14	440,273	2,140,984	1,281,886	1,155,674	3,664	22,043	53,739	54,4
E3	3,471,017	87.38	438,073	2,023,266	1,447,751	1,174,742	3,719	22,334	54,471	55,5
Overall	3,790,496	80.81	727,412	NA	NA	1,267,659	4,008	24,007	59,401	60,1

Table 5 Functional categories for coding SNPs （Download)

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Samples	Synonymous	Missense	Stopgain	Stoploss	Startloss	Splicing	Ti/Tv
622.5.041	17,503	11,014	160	35	11	145	2.88
752.33	17,296	11,183	175	28	8	150	3.04
799.36	17,006	10,467	139	32	9	101	2.84
805.044	17,459	11,035	164	33	11	142	2.87
814.292	17,349	10,640	151	35	6	109	2.87
E1	16,751	9,645	85	29	4	76	2.99
E3	17,014	9,780	88	32	4	77	3.00
Overall	17,928	11,657	184	35	11	158	2.82

5 InDel Results

There were totally 1,223,210 InDels called in all samples. Of these variants, 44.72% were represented in dbSNP and 676140.00% were annotated in the 1000 Genomes Project database. The number of novel InDels was NA. Of overall InDels , 378 were frameshift, 8 were stoploss, 8 were startloss and 96 were splice site(Table7). The summary statistics of InDels was showed in Table6. The length distribution of the InDels in coding sequence region(CDS) were also plotted as Figure5.

Table 6 Summary statistics for identified InDels (Download)

Samples	Total InDels	Fraction of InDels in dbSNP (%)	Novel	Homozygous	Heterozygous	Intronic	5' UTRs	3' UTRs	Upstream	Downs
622.5.041	1,098,900	50.24	546,765	567,642	531,258	400,659	787	7,706	21,342	19,305
752.33	1,047,729	50.78	515,688	663,723	384,006	381,576	758	7,431	20,408	18,547
799.36	1,061,713	50.07	530,140	591,110	470,603	389,400	764	7,546	20,901	18,990
805.044	1,097,157	50.24	545,898	569,812	527,345	400,660	760	7,685	21,370	19,336
814.292	1,094,810	50.47	542,208	572,902	521,908	399,055	786	7,709	21,222	19,236
E1	1,067,571	50.87	524,487	606,088	461,483	389,002	766	7,506	20,608	18,760
E3	1,079,335	51.27	525,991	574,620	504,715	394,955	787	7,642	20,897	19,072
Overall	1,223,210	44.72	676,140	NA	NA	446,416	849	8,522	23,971	21,522

Table 7 Functional categories for coding InDels (Download)

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Samples	Frameshift	Non-frameshift Insertion	Non-frameshift Deletion	Stoploss	Startloss	Splicing
622.5.041	299	161	198	8	8	77
752.33	289	149	185	7	5	78
799.36	312	151	209	8	6	85
805.044	301	164	203	8	7	78
814.292	318	162	206	9	5	84
E1	292	160	193	7	5	93
E3	298	159	193	7	6	88
Overall	378	172	220	8	8	96

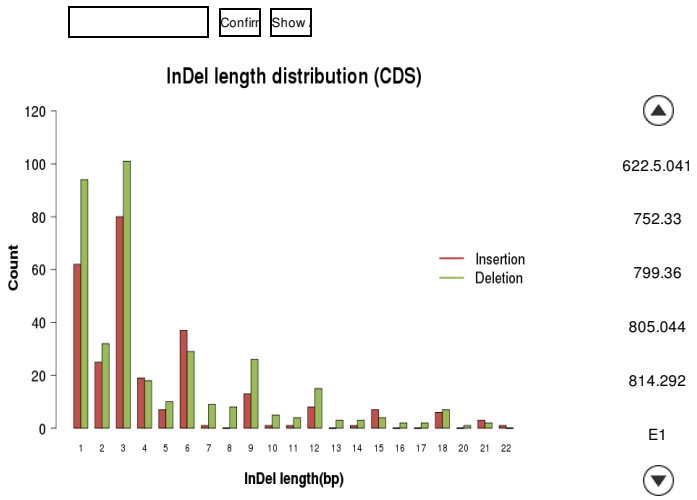


Figure 5 The distribution of lengths of coding InDel variants.

X-axis denotes the length of Insertions/Deletions, and Y-axis indicates the number of Insertions/Deletions.

Methods

1 Whole genome sequencing

The qualified genomic DNA sample was randomly fragmented by Covaris technology and the fragment of 350bp was obtained after fragment selection. The end repair of DNA fragments was performed and an "A" base was added at the 3'-end of each strand. Adapters were then ligated to both ends of the end repaired/dA tailed DNA fragments, then amplification by ligation-mediated PCR (LM- PCR), then single strand separation and cyclization. The rolling circle amplification (RCA) was performed to produce DNA Nanoballs (DNBs). The qualified DNBs were loaded into the patterned nanoarrays and pair-end read were read through on the BGISEQ-500 platform and high-throughput sequencing are performed for each library to ensure that each sample meet the average sequencing coverage requirement. Sequencing-derived raw image files were processed by BGISEQ-500 basecalling Software for base-calling with default parameters and the sequence data of each individual is generated as paired-end reads, which is defined as "raw data" and stored in FASTQ format.

2 Bioinformatics analysis overview

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Figure1 showed the data flow for the whole genome sequencing analysis.

The bioinformatics analysis began with the sequencing data (raw data from the BGISEQ machine). First, the clean data was produced by data filtering on raw data. All clean data of each sample was mapped to the mouse reference genome (mm10). Burrows-Wheeler Aligner (BWA)[1][2] software was used to do the alignment. To ensure accurate variant calling, we followed recommended Best Practices for variant analysis with the Genome Analysis Toolkit(GATK, https://www.broadinstitute.org/gatk/guide/best-practices). Local realignment around InDels and base quality score recalibration were performed using GATK[3][4], with duplicate reads removed by Picard tools[5]. The sequencing depth and coverage for each individual were calculated based on the alignments.

In addition, the strict data analysis quality control system(QC) in the whole pipeline was built to guarantee qualified sequencing data.

The genomic variations, including SNPs and InDels were detected by the state-of-the-art software, such as HaplotypeCaller of GATK(v3.3.0). After that, the variant quality score recalibration (VQSR) method, which uses machine learning to identify annotation profiles of variants that are likely to be real, was applied to get high-confident variant calls. The Copy Number Variants (CNVs) were called using the CNVnator[6] v0.2.7 read-depth algorithm. The structural variations (SV) were detected using Breakdancer[7][8] or CREST[9]. Then the SnpEff tool (http://snpeff.sourceforge.net/SnpEff_manual.html) was applied to perform a series of annotations for variants.

The final variants and annotation results were used in the downstream advanced analysis.

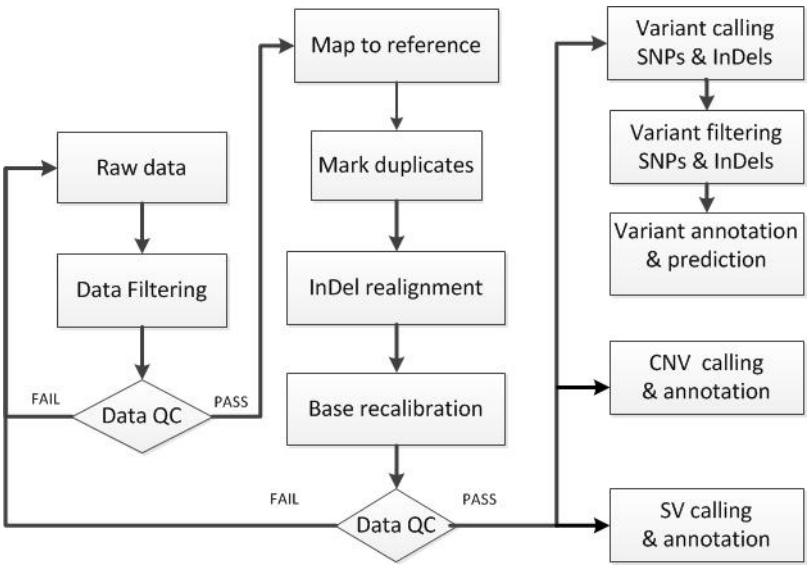


Figure 1 The whole genome sequencing analysis pipeline.

3 Data cleanup

In order to decrease noise of sequencing data, data filtering was done firstly, which included: (1) Removing reads containing sequencing adapter; (2) Removing reads whose low-quality base ratio (base quality less than or equal to 5) is more than 50%; (3) Removing reads whose unknown base ('N' base) ratio is more than 10%. Statistical analysis of data and downstream bioinformatics analysis

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were performed on this filtered, high-quality data, referred to as the " clean data ".

4 Mapping and marking duplicates

All clean reads were aligned to the mouse reference genome (mm10) using Burrows-Wheeler Aligner (BWA V0.7.12). We did mapping for each lane separately and also add the read group identifier, which by lane, into the alignment files. Here we used BWA-MEM method. Below are the BWA commands used for the alignments:

```
bwa mem -M -Y -R 'read_group_tag' ucsc.mm10.fasta read1.fq.gz read2.fq.gz |
samtools view -Sb - > aligned_reads. BAM
```

Here the 'read_group_tag' need to be provided, e.g., '@RG\tID:GroupID\tSM:SampleID\tPL:illumina\tLB:libraryID'.

Picard-tools(v1.118)[5] was used to sort the SAM files by coordinate and converted them to BAM files.

```
java -jar picard-tools-1.118/SortSam.jar I=aligned_reads. BAM
O=aligned_reads.sorted. BAM  SORT_ORDER=coordinate
```

The same DNA molecules can be sequenced several times during the sequencing process. The resulting duplicate reads are not informative and should not be counted as additional evidence for or against a putative variant. We used Picard tools(v1.118)[5] to mark the duplicate reads , which were ignored in downstream analysis.

```
java -jar picard-tools-1.118/MarkDuplicates.jar \
   I=aligned_reads.sorted. BAM  \
   O=aligned_reads.sorted.dedup. BAM  METRICS_FILE=metrics.txt \
   CREATE_INDEX=true
```

5 Local realignment around InDels

The realignment step identifies the most consistent placement of the reads relative to the InDel in order to clean up the artifacts. It occurs in two steps: first the program identifies intervals that need to be realigned, then in the second step it determines the optimal consensus sequence and performs the actual realignment of reads.

```
java -jar GenomeAnalysisTK.jar -T RealignerTargetCreator \
   -R gatk_ref/ucsc.mm10.fasta \
   -o indels_religner.intervals \

java -jar GenomeAnalysisTK.jar -T IndelRealigner \
   -R ucsc.mm10.fasta \
   -I aligned_reads.sorted.dedup. BAM  \
   -targetIntervals indels_religner.intervals \
   -o aligned_reads.sorted.dedup.realigned. BAM
```


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6 Base Quality Score Recalibration (BQSR)

The variant calling method heavily relied on the base quality scores in each sequence read. Various sources of systematic error from sequencing machines leaded to over- or under-estimated base quality scores. So the BQSR step was necessary to get more accurate base qualities, which in turn improved the accuracy of variant calls. The following commands were used to do this step.

```
java -jar GenomeAnalysisTK.jar -T BaseRecalibrator \
   -R gatk_ref/ucsc.hg19.fasta \
   -I aligned_reads.sorted.dedup.realigned. BAM  \
   -knownSites dbsnp_138.hg19. VCF  \
   -knownSites Mills_and_1000G_gold_standard. InDels .hg19. VCF  \
   -knownSites 1000G_phase1. InDels .hg19. VCF  \
   -o recal.table

java -jar GenomeAnalysisTK.jar -T PrintReads \
   -R gatk_ref/ucsc.hg19.fasta \
   -I aligned_reads.sorted.dedup.realigned. BAM  \
   -BQSR recal.table -o aligned_reads.sorted.dedup.realigned.recal. BAM
```

7 SNP and InDel calling

The HaplotypeCaller of GATK(v3.3.0) was used to call both SNPs and InDels simultaneously via local de-novo assembly of haplotypes in a region showing signs of variation. The raw variation set containing all potentially variants, which was outputted into the VCF file, was obtained by using this command.

```
java -jar GenomeAnalysisTK.jar -T HaplotypeCaller \
   -R gatk_ref/ucsc.hg19.fasta --genotyping_mode DISCOVERY \
   -I aligned_reads.sorted.dedup.realigned.recal. BAM  \
   -o raw_variants. VCF  -stand_call_conf 30 -stand_emit_conf 10 -minPruning
3
```

8 Variant filtering

When we obtained the raw variation set containing both SNPs and InDels , it is extremely important to apply filtering methods, in order to move on to downstream analyses with the highest-quality call set possible. We used the GATK Variant Quality Score Recalibration (VQSR) that uses machine learning algorithm to filter the raw variant callset. The GATK VQSR used high-quality known variant sets as training and truth resources and built a predictive model to filter spurious variants. The SNPs and InDels marked PASS in the output VCF file were high-confident variation set.

For SNPs recalibration strategy, we used the following datasets and features to train the model. (a) Training sets: HapMap V3.3, Omni2.5M genotyping array data and high-confidence SNP sites produced by the 1000 Genomes Project. (b) Features: Coverage (DP), Quality/depth (QD), Fisher test on strand bias(FS), Odds ratio for strand bias (SOR), Mapping quality rank sum test (MQRankSum), Read position rank sum test (ReadPosRankSum), RMS mapping quality (MQ).

The recalibration commands and parameters for SNPs were the following.

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```
java -jar GenomeAnalysisTK.jar -T SelectVariants \
   -R gatk_ref/ucsc.hg19.fasta \
   -V raw_variants. VCF  -selectType  SNP  \
   -o raw_snps. VCF

java -jar GenomeAnalysisTK.jar  -T VariantRecalibrator \
   -R gatk_ref/ucsc.hg19.fasta -input raw_snps. VCF  \
   -resource:hapmap,known=false,training=true,truth=true,prior=15.0
hapmap_3.3.hg19. VCF  \
   -resource:omni,known=false,training=true,truth=true,prior=12.0
1000G_omni2.5.hg19. VCF  \
   -resource:1000G,known=false,training=true,truth=false,prior=10.0
1000G_phase1. SNPs .high_confidence.hg19. VCF  \
   -resource:dbsnp,known=true,training=false,truth=false,prior=2.0
dbsnp_138.hg19. VCF  \
   -an DP -an QD -an FS -an SOR -an MQ -an MQRankSum -an ReadPosRankSum \
   -mode  SNP  \
   -tranche 100.0 -tranche 99.9 -tranche 99.0 -tranche 90.0 \
   -recalFile recalibrate_SNP.recal \
   -tranchesFile  recalibrate_SNP.tranches \
   -rscriptFile recalibrate_SNP_plots.R

java -jar GenomeAnalysisTK.jar -T ApplyRecalibration \
    -R gatk_ref/ucsc.hg19.fasta \
    -input raw. SNP . VCF  \
    -mode  SNP  \
    --ts_filter_level 99.0 \
    -recalFile recalibrate_SNP.recal \
    -tranchesFile recalibrate_SNP.tranches \
    -o filtered_snp. VCF
```

For InDels recalibration strategy, we used the following datasets and features to train the model. (a) Training sets: Mills 1000G gold standard InDel set. (b) Features: Coverage (DP), Quality/depth (QD), Fisher test on strand bias(FS), Odds ratio for strand bias (SOR), Mapping quality rank sum test (MQRankSum), Read position rank sum test (ReadPosRankSum).

The recalibration commands and parameters for InDels were the following.

```
java -jar GenomeAnalysisTK.jar -T SelectVariants \
   -R gatk_ref/ucsc.hg19.fasta \
   -V raw_variants. VCF  -selectType  InDel  \
   -o raw_indels. VCF

java -jar GenomeAnalysisTK.jar -T VariantRecalibrator \
   -R gatk_ref/ucsc.hg19.fasta -input raw_indels. VCF  \
   -resource:mills,known=true,training=true,truth=true,prior=12.0
Mills_and_1000G_gold_standard. InDels .hg19. VCF
```


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```
   -an QD -an DP -an FS -an SOR -an MQRankSum -an ReadPosRankSum -mode
 InDel  \
   -tranche 100.0 -tranche 99.9 -tranche 99.0 -tranche 90.0 --maxGaussians 4
\
   -recalFile recalibrate_INDEL.recal \
   -tranchesFile recalibrate_INDEL.tranches  \
   -rscriptFile recalibrate_INDEL_plots.R

java -jar GenomeAnalysisTK.jar -T ApplyRecalibration \
   -R gatk_ref/ucsc.hg19.fasta \
   -input raw_indels. VCF  \
   -mode  InDel  \
   --ts_filter_level 99.0 \
   -recalFile recalibrate_INDEL.recal \
   -tranchesFile recalibrate_INDEL.tranches  \
   -o filtered_indel. VCF
```

9 Copy number variant calling

The Copy Number Variants (CNVs) were called using the CNVnator[6] v0.2.7 read-depth algorithm. The algorithm divides the genome into non-overlapping bins of equal size and uses the count of mapped reads in each bin as the Read-Depth signal. We used standard settings and a bin size of 100bp. The running involved few steps below.

```
cnvnator -root out.root -tree sample. BAM  -unique
cnvnator -root out.root -his 100 -d hg19_chr_fa_dir
cnvnator -root out.root -stat 100
cnvnator -root out.root -partition 100
cnvnator -root out.root -call 100 > sample. CNV
```

10 Structural variant calling

We provided genome-wide detection of five types of structural variants: inter-chromosomal translocations (CTX), intra-chromosomal translocations (ITX), inversions (INV), deletions (DEL), and insertions (INS).

Breakdancer[7][8] with default settings was used to detect structural variations (SV). This method implemented a paired-end discordance mapping algorithm based on the separation distance and alignment orientation between paired reads.

CREST[9] was used to identify SVs with standard settings. It mapped the breakpoints of SVs by using the information of soft-clipping reads and applying an assembly-mapping-searching-assembly-alignment procedure consisting of CAP3[10] and BLAT[11].

The following commands were used to call SVs .

```
breakdancer_max sample.cfg >sample.out
extractSClip.pl -i sample. BAM  --ref_genome ucsc.hg19.fasta -o outdir
```


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```
CREST.pl -f outdir/sample. BAM .cover -d sample. BAM  --ref_genome
ucsc.hg19.fasta -t ucsc.hg19.fasta.2bit --cap3 /path/to/cap3 --blatclient
/path/to/gfClient --blat /path/to/blat -o outdir
```

11 Variant annotation and prediction

After high-confident variants were identified, the SnpEff tool (http://snpeff.sourceforge.net/SnpEff_manual.html) was applied to perform:

(a) gene-based annotation: identify whether variants cause protein coding changes and the amino acids that are affected.

(b) filter-based annotation: identify variants that are reported in dbSNP v141, or identify the subset of variants with MAF <1% in the 1000 Genome Project, or identify subset of coding non-synonymous SNPs with SIFT score<0.05, or find intergenic variants with GERP++ score>2, or many other annotations on specific mutations.

12 Web Resources

The URLs for data presented herein and data format details are as follows:

UCSC build mm10, http://hgdownload.cse.ucsc.edu/goldenPath/mm10/bigZips

RefGene database, http://hgdownload.cse.ucsc.edu/goldenPath/mm10/Database/refGene.txt.gz

dbSNP, http://www.ncbi.nlm.nih.gov/snp

SAM / BAM file format, Sequence Alignment/Map Format Specification http://samtools.github.io/hts-specs/SAMv1.pdf

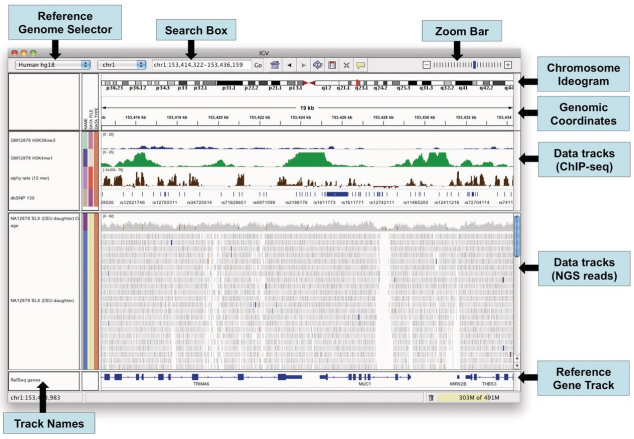
VCF format, http://www.1000genomes.org/wiki/analysis/vcf4.0

Help

1 Guide to visualization

The Integrative Genomics Viewer (IGV) is a high-performance visualization tool for interactive exploration of many different types of large genomic datasets. IGV is freely available for download from http://www.broadinstitute.org/igv. IGV includes a large number of specialized features for exploring next-generation sequencing read alignments, including features for sequencing coverage and variant visualization. IGV supports SAM / BAM read alignment file formats and VCF format for viewing variants. The following figure illustrated the IGV application window.

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2 Guide to selecting variants for validation

When we obtained variant call sets, we maybe want to select some interesting variants for validation using other platforms such as Sanger sequencing, Sequenom MassARRAY and array-based platform. With the help of IGV visualization, we can choose the target variants by hand. Here is a suggestion. DO NOT choose variants with the following features:

(1) variants neighboring InDels called.

(2) variants located in tandem repeat sequence regions.

(3) variants located in homologous sequence regions. UCSC BLAT tool can be used to find areas of probable homology. From the reference genome, we can obtain a query sequence by extending 100bp towards both sides of the variant and submit the sequence to http://genome.ucsc.edu/cgi-bin/hgBlat\?command=start.

(4) heterozygous variants with sequencing allele unbalance, namely the fraction of reads supporting alternate allele is less than 0.25 or more than 0.75.

3 Guide to finding candidate variants

When we want to find candidate variants, we can use the variant annotation results and focus only on non-synonymous variants, splicing mutations and frameshift coding insertions/deletions. (1)Remove variants with MAF >=1% according to allele frequency from the 1000 Genomes Project control database. (2)Remove variants with MAF >=1% according to allele frequency of European American population from NHLBI-ESP6500 control database.(3)Remove variants with MAF >=1% according to allele frequency of African American population from NHLBI-ESP6500 control database. (4)Report the putative pathogenicity of variants. Use SIFT/PolyPhen2/Mutation assessor/Condel/FATHMM scores to predict whether a variant and an amino acid substitution affects protein function. If SIFT score<=0.05 or PolyPhen2>=0.909 or MA score>=1.9 or Condel = deleterious or FATHMM=deleterious, we predict this variant as a deleterious variant.

4 Format of annotation files

Format of SNP annotation file

Format of InDel annotation file

5 Decompress the file

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All the data were compressed as file format of *.tar.gz by "tar -czvf" under linux environment. Please decompress them as follows:

Unix/Linux user: tar -zxvf *.tar.gz.

Windows user: 'winRAR' is recommended.

Mac user: shell: tar -zxvf *.tar.gz, and 'stuffit expander' is recommended.

FAQs

How to view BAM files in Microsoft Windows ?

Create index of BAM using Picard tools, named *.bai. Then open it with IGV.

Why do we use BWA-MEM?

BWA-MEM is designed for longer sequences ranged from 70bp to 1Mbp and split alignment. It is also the latest and is generally recommended for high-quality queries as it is faster and more accurate. The performance evaluation of BWA-MEM can be seen in the related paper (Li H. 2013, arXiv) http://arxiv.org/pdf/1303.3997v2.pdf.

What's the fragment length range of small InDel in exome and whole genome re-sequencing?

For small InDel , the range is from 1 to 50bp.

Base quality is not completely true, do we take this situation into consideration in variant calling?

Yes. Base Quality Score Recalibration step was used to correct raw base quality scores before calling variants by GATK package.

Do we use UnifiedGenotyper or HaplotypeCaller to call variants in GATK v3.3.0 in the pipeline?

Use HaplotypeCaller. The HaplotypeCaller is a more recent and sophisticated tool than the UnifiedGenotyper. Its ability to call SNPs is equivalent to that of the UnifiedGenotyper, its ability to call InDels is far superior, and it is now capable of calling non-diploid samples.

Compared to whole genome re-sequencing, exome sequencing only for the exon regions of DNA can be more simple, economical and efficient. Why should we select whole genome re-sequencing? And what's the sense?

The large structure variations and the mutations in non-exome region can be calling by whole genome sequencing, so that we will have a more comprehensive understanding of genome.

What is the purpose of checking genotype barcode of 21 SNP sites in data quality control step?

Genotype barcode of 21 SNP sites should be checked with sequencing data calls. We generate genotype barcode of 21 SNP sites by Sequenom for all the samples to track the identity of samples during the sequencing process.

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